

REPORT

ARTIFICIAL INTELLIGENCE (CSA1717)

Assessment Tool 3 – Dry Run Test

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1.INTRODUCTION

Artificial Intelligence (AI) enables computers to solve problems and make intelligent decisions efficiently. Search and decision-making algorithms are important techniques used in AI to find optimal solutions. The A* Search Algorithm is an informed search algorithm that finds the shortest path by using the path cost $g(n)$ and heuristic value $h(n)$. The Minimax Algorithm with Alpha-Beta Pruning is used in two-player games to determine the best move while

reducing unnecessary node evaluations through pruning. In this report, both algorithms are implemented to demonstrate efficient pathfinding and optimal decision-making in Artificial Intelligence.

2.DRY RUN OF A SEARCH ALGORITHM

Given:

Edge Costs

- $A \rightarrow B = 2$
- $A \rightarrow C = 4$
- $B \rightarrow C = 3$
- $B \rightarrow D = 7$
- $B \rightarrow E = 2$
- $C \rightarrow E = 3$
- $D \rightarrow E = 2$
- $D \rightarrow G = 2$

Heuristic Values

Node $h(n)$

A	7
B	6
C	4
D	3
E	2
G	0

Step 1

Current Node = A

- $g(A) = 0$
- $h(A) = 7$

- $f(A) = 7$

Expand A:

Node	$g(n)$	$h(n)$	$f(n)$
B	2	6	8
C	4	4	8

Open List: B(8), C(8)

Closed List: A

Step 2

Choose **B** (tie broken alphabetically).

Expand B.

Possible paths:

- $C = 2 + 3 = 5$ (Existing cost to C is 4, so ignore)
- $D = 2 + 7 = 9$
- $E = 2 + 2 = 4$

Node $g(n)$ $h(n)$ $f(n)$

C 4 4 8

E 4 2 6

D 9 3 12

Open List: E(6), C(8), D(12)

Closed List: A, B

Step 3

Choose **E** (lowest f).

Expand E.

No direct edge to G.

Open List: C(8), D(12)

Closed List: A, B, E

Step 4

Choose **C**.

Expand C.

New path to E:

$$g(C) + 3 = 7$$

$$\text{Existing } g(E) = 4$$

Ignore (higher cost).

Open List: D(12)

Closed List: A, B, E, C

Step 5

Choose **D**.

Expand D.

Path to G:

$$g(D) = 9$$

$$g(G) = 9 + 2 = 11$$

Node g(n) h(n) f(n)

G 11 0 11

Open List: G(11)

Closed List: A, B, E, C, D

Step 6

Choose **G**.

Goal reached.

Open List: Empty

Closed List: A, B, E, C, D, G

Iteration	Current Node	Open List	Closed List
1	A	B(8), C(8)	A
2	B	E(6), C(8), D(12)	A, B
3	E	C(8), D(12)	A, B, E
4	C	D(12)	A, B, E, C
5	D	G(11)	A, B, E, C, D
6	G	Empty	A, B, E, C, D, G

Optimal Path

A → B → D → G

Total Path Cost

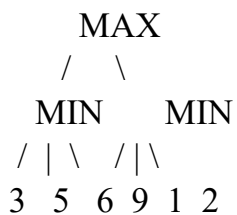
$$2 + 7 + 2 = \boxed{11}$$

Answer:

- **Optimal Path: A → B → D → G**
- **Total Path Cost: 11**

3.DRY RUN OF MINIMAX WITH ALPHA-BETA PRUNING

Given Game Tree



Evaluate left to right.

Initial values:

- $\alpha = -\infty$

- $\beta = +\infty$

Step 1: Root (MAX)

MAX

$$\alpha = -\infty$$

$$\beta = +\infty$$

Go to the left MIN node.

Step 2: Left MIN Node

Initial:

$$\alpha = -\infty$$

$$\beta = +\infty$$

Evaluate children.

Visit 3

$$\beta = \min(+\infty, 3) = 3$$

Current:

$$\alpha = -\infty$$

$$\beta = 3$$

Since

$$\beta(3) \leq \alpha(-\infty)?$$

No.

Visit 5

$$\beta = \min(3, 5) = 3$$

No pruning.

Visit 6

$$\beta = \min(3, 6) = 3$$

Left MIN returns

$$\text{Value} = 3$$

Step 3: Back to Root (MAX)

Current value

$$\text{MAX} = \max(-\infty, 3) = 3$$

Update

$$\alpha = 3$$

$$\beta = +\infty$$

Go to the right MIN node.

Step 4: Right MIN Node

Initial

$$\alpha = 3$$

$$\beta = +\infty$$

Visit 9

$$\beta = \min(+\infty, 9) = 9$$

Check

$$9 \leq 3 ?$$

No.

Visit 1

$$\beta = \min(9, 1) = 1$$

Now

$$\beta = 1$$

$$\alpha = 3$$

Since

$$\beta \leq \alpha$$

$$1 \leq 3$$

YES

Alpha-Beta Cutoff occurs.

So node 2 is pruned.

Right MIN returns

$$\text{Value} = 1$$

Step 5: Back to Root

$$\text{MAX} = \max(3, 1) = 3$$

Final value

3

DRY RUN TABLE

Step	Current Node	α	β	Selected Value	Pruned
1	MAX	$-\infty$	$+\infty$	—	—
2	Left MIN	$-\infty$	3	3	None
3	MAX	3	$+\infty$	3	—
4	Right MIN	3	1	1	2
5	MAX	3	$+\infty$	3	—

FINAL ANSWER

Best Move for MAX

Choose Left MIN

Final Minimax Value

3

Pruned Node

2

4.CONCLUSION

The implementation of the A* Search Algorithm and Minimax with Alpha-Beta Pruning demonstrates two important techniques used in Artificial Intelligence for problem-solving and decision-making. The A* algorithm efficiently finds the shortest path using heuristic information, while the Minimax algorithm with Alpha-Beta Pruning determines the optimal move by eliminating unnecessary node evaluations. Both algorithms improve efficiency, reduce computational effort, and provide optimal solutions, making them widely applicable in real-world AI applications such as navigation systems, robotics, and game-playing systems.

5.REFERENCES

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